$$\begin{vmatrix} 1-\lambda & -2 \\ -2 & 5-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)(5-\lambda) - 4 = 0$$

$$\Rightarrow 5 - 6\lambda + \lambda^2 - 4 = 0$$

$$\Rightarrow \lambda^2 - 6\lambda + 1 = 0$$

$$\Rightarrow \lambda = \frac{6 \pm \sqrt{36-4}}{2} = \frac{6 \pm \sqrt{32}}{2}$$

Thm:-

Let A be a real positive definite matrix, then the function $\langle u, v \rangle = u^T A v$, then the fun^n is an inner product on $\mathbb{R}^n$.

Matrix Representation of an innerproduct

Let V be an innerproduct space with the basis, $S = \{u_1, u_2, \ldots, u_n\}$, then the matrix $A = [a_{ij}]$, $i,j = 1,2,\ldots,n$ where $a_{ij} = \langle u_i, u_j \rangle$ is called the matrix representation of an innerproduct in V relative to the basis S.

Ex:- $S = \{1, t, t^2\}$, $V = P_2$, $[-1, 1]$

$$\langle f, g \rangle = \int_{-1}^{1} f(t) g(t)\, dt$$

Find the matrix representation of S in V.

P Kalika Maths